Abstract

This study investigates patterns in national renewable electricity shares using the Our World in Data energy dataset. A cleaned panel of 2,191 country-year observations from 212 countries supports descriptive trend analysis, unsupervised grouping of electricity-generation profiles, and one-year-ahead prediction of the renewable share of electricity. Country-level K-Means clustering identifies five groups (silhouette score 0.501) among 207 countries, although latest available reporting years differ. For prediction, 1,439 training examples and 279 held-out examples are separated by target year. The persistence baseline attains the lowest test mean absolute error (MAE 1.965 percentage points), outperforming Random Forest (2.144) and histogram gradient boosting (2.159). The result emphasizes the importance of meaningful baselines, temporal evaluation and transparent reporting over model complexity.

1. Research objectives and questions

The project has three related aims: (i) describe the observed evolution of country-average renewable electricity shares; (ii) characterize differences in solar, wind and hydropower electricity shares through country clustering; and (iii) assess whether nonlinear supervised models improve next-year renewable-share forecasts relative to a persistence rule.

Research questions. How does the unweighted average across available countries evolve over the study window? Which renewable generation profiles emerge from clustering? Do Random Forest or gradient boosting outperform the last-observed-value baseline on later target years? How do errors differ across years and countries?

2. Data acquisition and audit | Notebook Steps 1–3

Step 1 — Data ingestion

The notebook reads the OWID energy CSV directly from the project's GitHub raw-data endpoint using pandas.read_csv. The saved run reports 23,377 records, 130 columns and a maximum reported year of 2025. The source is a living dataset, so a later rerun may produce different totals unless a specific release or file snapshot is retained.

Step 2 — Feature screening and missingness

Ten candidate columns are checked: country, year, ISO code, renewable electricity share, solar share, wind share, hydro share, electricity demand, GDP and population. The notebook explicitly prints column availability and null counts before cleaning. For example, the original table has 15,505 missing values in renewable electricity share and 11,597 in GDP. These diagnostics justify a narrower, target-focused analytical panel.

Step 3 — Cleaning and panel construction

The cleaning logic retains three-letter ISO country codes and years 2015–2025, drops records without the renewable electricity share target, removes duplicate country-year entries, and sorts observations chronologically within country. The resulting dataset has 2,191 records from 212 countries. Solar and renewable electricity shares have no missing values in this panel; wind and hydro retain 38 and 35 missing values, respectively, while GDP retains 871. The notebook does not use GDP in the predictive feature set.

STAGE	NOTEBOOK OUTPUT
Source dataset	23,377 records × 130 variables
Filtered and cleaned panel	2,191 records; 212 countries
Panel window	2015–2025
Missing wind / hydro in cleaned panel	38 / 35 records

STAGE	NOTEBOOK OUTPUT
Missing GDP in cleaned panel	871 records

3. Exploratory analysis | Step 4

The notebook groups the cleaned panel by year and computes the arithmetic mean across available country observations of renewables_share_elec. It then plots this annual country average. The displayed average rises from 30.80% in 2015 to 44.65% in 2025, a difference of 13.85 percentage points.

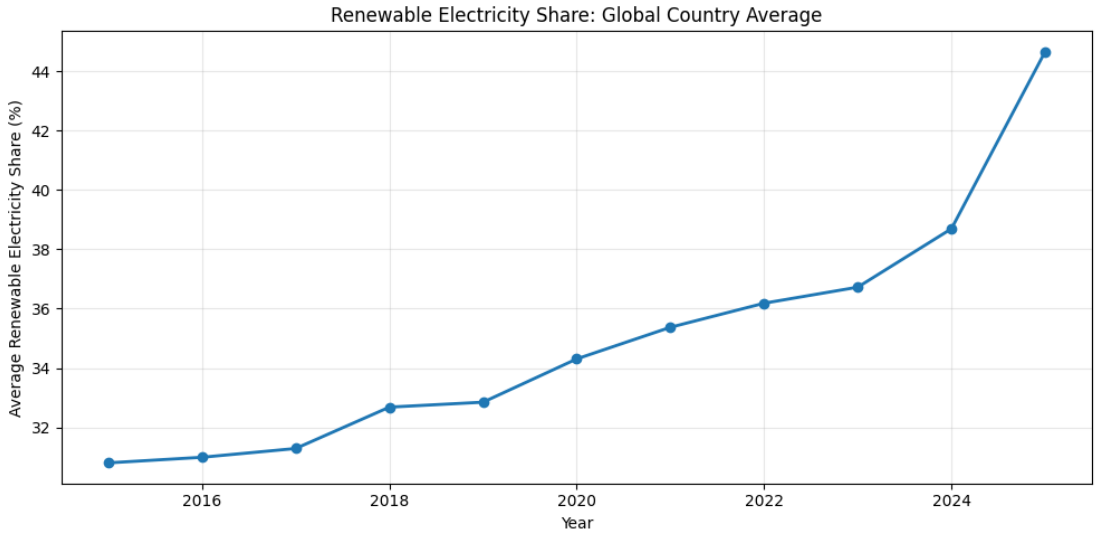


Figure 1. Country-average renewable electricity share by year, reproduced from the notebook. The measure is unweighted across available countries; it is not the share of total world electricity generated from renewables.

Interpretation. The plotted series is evidence of a higher mean among countries observed in the dataset. It should not be presented as a globally generation-weighted renewable share. Country coverage may change by year; therefore, changes in the series can reflect both within-country movements and changes in which countries are represented.

4. Unsupervised learning | Step 5

For each country, the notebook selects the latest available row, removes rows with missing solar, wind or hydro electricity shares, and standardizes these three features using StandardScaler. K-Means models are fit for k = 2, 3, 4, 5 and 6 with random_state = 42 and n_init = 10. The model with the highest silhouette score is selected. The saved result chooses five clusters for 207 countries, with a silhouette score of 0.501.

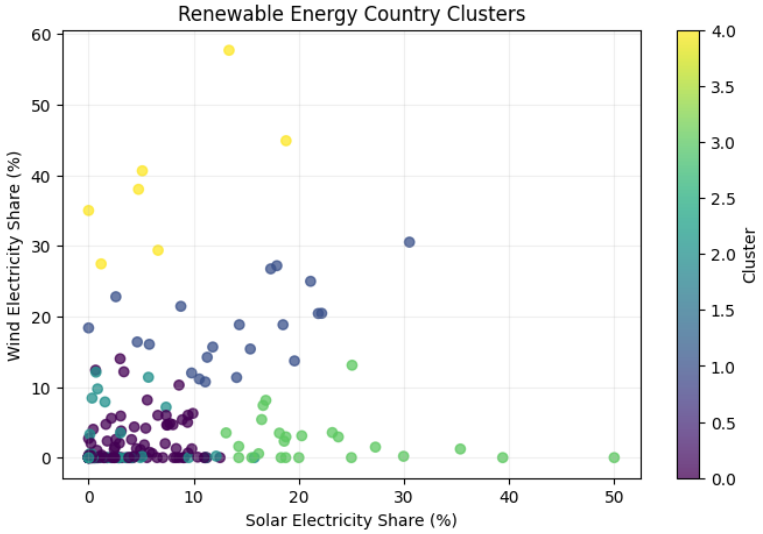


Figure 2. Country clusters projected onto solar and wind shares, reproduced from the notebook. Hydropower share is included in clustering but is not directly visible on the two-dimensional axes.

5. Cluster interpretation | Steps 5, 5a and 14

The final notebook reports mean and median shares within each cluster, the number of countries per cluster, and the reporting-year distribution. Cluster IDs are arbitrary model labels; descriptions below are interpretations of the observed centroids rather than official country classifications.

CLUSTER	N	SOLAR MEAN	WIND MEAN	HYDRO MEAN
0	110	3.45%	1.58%	8.08%
1	21	13.77%	18.43%	16.50%
2	45	2.32%	1.45%	74.24%
3	24	22.30%	2.54%	12.70%
4	7	7.12%	39.00%	7.19%

Interpretive profiles

Cluster 0 — lower shares in the three selected technologies: 110 countries, with mean solar, wind and hydro shares of 3.45%, 1.58% and 8.08%. This is a description of the selected variables only; it does not establish a country’s total energy transition status.

Cluster 1 — diversified selected renewable mix: 21 countries, with comparatively material contributions from all three technologies (13.77% solar, 18.43% wind, 16.50% hydro).

Cluster 2 — hydro-dominant: 45 countries; mean hydropower electricity share reaches 74.24%, substantially higher than the other clusters.

Cluster 3 — solar-oriented: 24 countries; the highest cluster-average solar share at 22.30%, with more modest mean wind and hydro shares.

Cluster 4 — wind-oriented: 7 countries; the highest cluster-average wind share at 39.00%. The small group size warrants caution in generalizing the profile.

Reporting-year comparability

LATEST OBSERVATION YEAR	COUNTRIES
2022	2
2023	14
2024	103
2025	88

The country profiles represent the latest available observation rather than a single synchronized cross-section. Comparing a country observed in 2022 with one observed in 2025 introduces time-related variation. The cluster labels should therefore be read as descriptive profiles, not as time-controlled rankings or causal categories.

6. Forecast design | Step 6

The supervised task predicts a country's next-year renewable electricity share from the current-year share, previous-year share and current-year solar, wind and hydro shares. Within each country, pandas groupby and shift construct the previous and next observations. The code retains a modeling row only when the preceding and following records correspond to exactly adjacent years, and it drops rows missing any model feature or the next-year target.

MODEL INPUT	INTERPRETATION
renewables_share_elec	Renewable electricity share at year t
previous_share	Renewable electricity share at year t−1
solar_share_elec	Solar electricity share at year t
wind_share_elec	Wind electricity share at year t
hydro_share_elec	Hydropower electricity share at year t
next_share (target)	Renewable electricity share at year t+1

The training partition contains 1,439 rows whose target years are 2017–2023. The held-out test partition contains 279 rows whose target years are 2024–2025. This is a chronological split on the year of the predicted outcome, not a random row split. The target is not included as an input feature.

Baseline and learning algorithms

Persistence baseline: predicts next-year share as equal to the current-year share. Random Forest: RandomForestRegressor with 100 trees, maximum depth 10, random_state 42 and n_jobs = −1. Gradient Boosting: HistGradientBoostingRegressor with 150 maximum iterations, learning rate 0.05, 15 maximum leaf nodes, L2 regularization 1.0 and random_state 42. No hyperparameter search or cross-validation is implemented in the final notebook; these are the specified configurations.

7. Model evaluation | Steps 7–8

The notebook computes mean absolute error (MAE) on the same held-out test set for all three predictors. MAE is the average absolute difference between actual and predicted renewable electricity shares, measured in percentage points. The notebook also reports R^2 of approximately 0.989 for both learned models; R^2 alone is not sufficient to establish superiority over the baseline.

METHOD	TEST MAE	COMPARISON
Persistence baseline	1.965	Best overall
Random Forest	2.144	0.179 points above baseline
Histogram Gradient Boosting	2.159	0.194 points above baseline

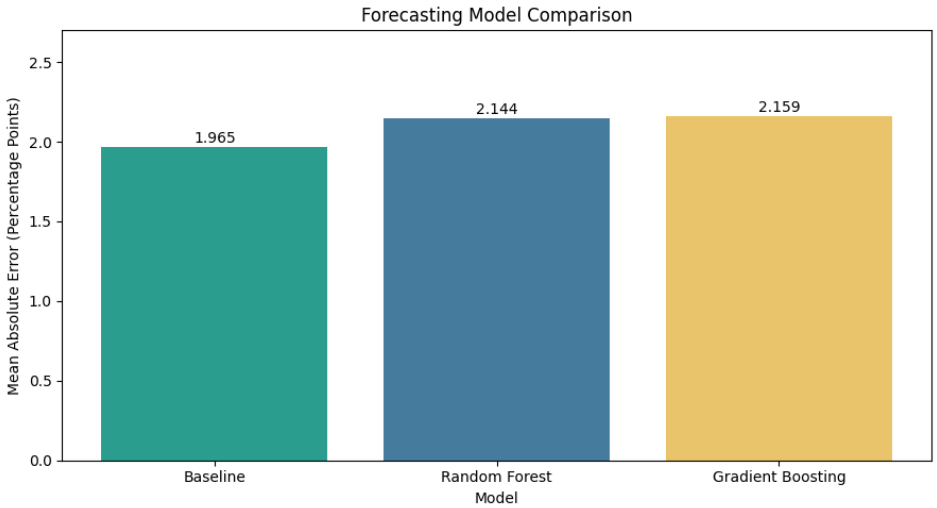


Figure 3. Test-set MAE for the persistence baseline and two machine-learning models. Lower values indicate better average absolute predictive accuracy. Original notebook figure.

8. Temporal integrity and validation | Steps 9–10

The temporal audit scans the cleaned country panel for adjacent-year continuity. The notebook reports 2,191 cleaned rows, 1,767 rows with consecutive observations on both sides, zero detected year gaps and 1,718 modeling rows before the train/test split. It verifies that every modeled row has consecutive neighboring years.

The final validation cell independently checks that training target years (2017–2023) do not overlap test target years (2024–2025), that the target variable is absent from X_train, and that no model-feature values are missing in the training or testing matrices. The saved output states: “Basic validation checks passed.”

Scope of this check. These assertions provide evidence for temporal consistency and basic target exclusion; they do not constitute a comprehensive guarantee against every possible form of leakage, dataset revision, or selection bias. Because the source CSV is loaded from a moving URL, an exact replication also requires a pinned dataset snapshot.

9. Model interpretation | Step 11

Random Forest impurity-based feature importances show that the current renewable electricity share dominates the fitted model. The result is consistent with strong year-to-year persistence, but impurity importance is model-specific and should not be interpreted as a causal effect or a direct estimate of incremental predictive gain.

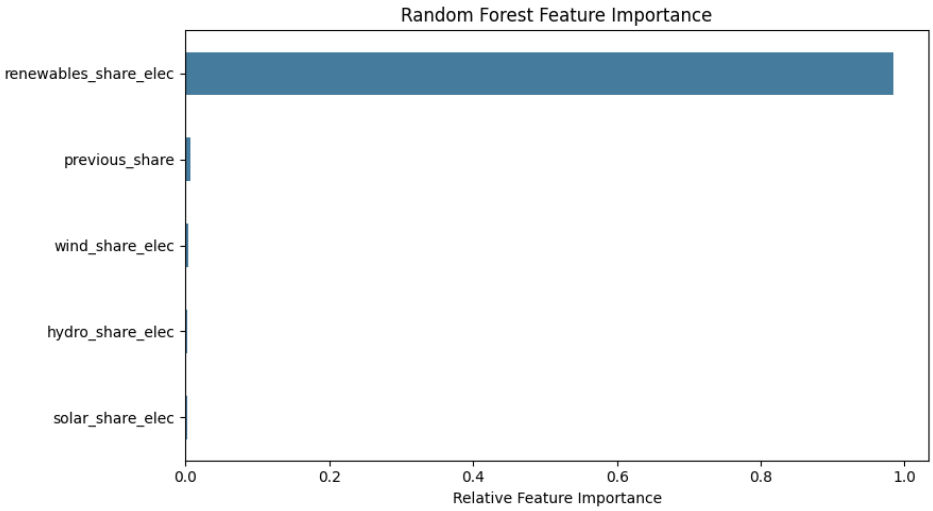


Figure 4. Random Forest relative feature importance, reproduced from the notebook.

FEATURE	IMPORTANCE
Current renewable share	0.9840
Previous renewable share	0.0075
Wind share	0.0033
Hydro share	0.0026
Solar share	0.0025

10. Country-level error analysis | Step 12

The notebook calculates absolute forecast errors by country and aggregates each country's baseline and Random Forest MAE. Among 191 evaluated countries, Random Forest has lower country-level MAE in 84, while persistence has lower MAE in 107. The distribution reveals heterogeneity that a single global test score cannot capture.

COUNTRY	TEST ROWS	BASELINE MAE	RF MAE
Nauru	1	20.000	18.847
Colombia	2	10.666	8.729
Macao	1	2.564	8.518
Lebanon	1	9.744	8.096
Montenegro	2	8.744	8.050
Chile	2	5.284	7.816
North Macedonia	2	7.569	6.508
Venezuela	1	0.797	6.474
Liberia	1	2.991	6.352
Sri Lanka	2	6.528	6.343

These are the ten countries with the highest Random Forest MAE in the notebook output. Several have only one held-out observation. Such estimates are unstable and should be treated as diagnostic case examples rather than reliable long-run country rankings.

11. Year-by-year forecast stability | Step 13

The test period is separated into 2024 and 2025 target years. Persistence has the lowest MAE in both years; all three models show higher errors in 2025. Because the second year contains fewer evaluated records, the difference is descriptive and does not establish a general trend in model deterioration.

TARGET YEAR	N	PERSISTENCE	RANDOM FOREST	GRADIENT BOOSTING
2024	191	1.767	1.938	1.897
2025	88	2.395	2.592	2.728

12. Integrated findings and discussion

Finding 1 — The descriptive country average increases

The notebook’s unweighted country average rises from 30.80% to 44.65% between the endpoints of its 2015–2025 panel. This finding describes the countries represented each year and should not be reworded as a generation-weighted global transition rate.

Finding 2 — Renewable electricity profiles are heterogeneous

Five clusters separate profiles associated with hydropower dominance, stronger solar orientation, stronger wind orientation, a more mixed renewable portfolio, and relatively low shares across the three selected technologies. The silhouette score of 0.501 indicates some separation under the selected standardized features, not proof that the groups are uniquely correct.

Finding 3 — Forecast persistence is a difficult benchmark

The baseline beats both fitted models overall and in each reported test year. Current renewable share also dominates Random Forest importance. Together, these observations suggest that short-horizon stability is a substantial predictive signal in this dataset. They do not imply that machine learning is never useful; only that these particular configurations did not outperform persistence under the notebook’s evaluation design.

Finding 4 — Aggregate metrics conceal local variation

Country-level evaluation shows that the forest wins in 84 countries despite losing overall and in a majority of the 191 country summaries. The combination of heterogeneous national profiles and sparse test records motivates careful interpretation rather than a single universal claim about forecast quality.

13. Methodological limitations

Dataset revision risk. The source is downloaded from a continuously updated branch, without a pinned version. Changing country coverage. The EDA trend is unweighted and may be affected by availability. Asynchronous clustering years. The latest available country records span 2022–2025. Limited holdout horizon. Only two target years are evaluated, with 88 records in the later year. Restricted predictors. GDP, demand and other possible explanatory factors are not used in forecasting. Model selection. No rolling-origin backtest or systematic hyperparameter optimization is documented. Interpretability. Impurity importance is not a causal or independent contribution measure.

14. Conclusion

This notebook demonstrates a complete, interpretable Data Science workflow: data ingestion and quality checks, structured cleaning, descriptive visualization, unsupervised segmentation, temporally separated prediction, baseline benchmarking, validation and subgroup error analysis. Its central forecasting conclusion is deliberately conservative: the simple persistence benchmark outperforms the two more complex models on the held-out observations. The study is most useful as an evidence-based comparison of methods and renewable electricity profiles, not as a claim of improved forecasting accuracy through model complexity.